

Operations with vectors in coordinates; applications

Everything from the last four lessons, applied to real proofs and real journeys.

LESSON 44–45

2 × 40 MIN

GEOMETRY 8, ТЕМЫ 40–41

STAGE 9 · 13.1–13.2

LESSON CLOCK



Recall the four operations 6 min

Position vectors 12 min

Vector proofs 12 min

Journeys and velocities 6 min

Homework 4 min

LEARNING OBJECTIVES

- Combine addition, subtraction and scalar multiplication in coordinates.
- Use position vectors to find midpoints and points dividing a segment.
- Prove geometric statements with vectors.
- Solve displacement and velocity problems.

TEXTBOOK

National Темы 40–41, pp. 90–95

Cambridge Learner's Book pp. 284–296

Workbook Workbook 13.2

01 Explanation

Position vectors

The **position vector** of A is OA , drawn from the origin. If $A(x; y)$ then $OA = (x; y)$ — a point and its position vector carry the same two numbers. From that one identification everything follows:

$$AB = OB - OA \mid OM = \frac{1}{2}(OA + OB) \text{ for the midpoint } M \text{ of } AB$$

To reach a point P that divides AB in a given proportion, start at A and travel part of the way:

$$OP = OA + t \cdot AB, \text{ where } t = 0 \text{ is } A \text{ and } t = 1 \text{ is } B$$

$t = \frac{1}{2}$ gives the midpoint, $t = \frac{1}{3}$ gives the point one third of the way, and so on.

Proving with vectors

A vector proof has three moves, and only three:

1. Name two vectors as ***a*** and ***b*** — usually two sides from one vertex.
2. Write every other vector in the figure in terms of them.
3. Read the conclusion: a multiple means **parallel**; equal vectors mean **equal and parallel**; **0** means the path closes.

THE MIDLINE THEOREM, IN THREE LINES

Let $AB = \mathbf{c}$ and $AC = \mathbf{b}$, with M and N the midpoints of AB and AC .

$$MN = AN - AM = \frac{1}{2}\mathbf{b} - \frac{1}{2}\mathbf{c} = \frac{1}{2}(\mathbf{b} - \mathbf{c}) = \frac{1}{2}BC$$

A multiple of BC , so $MN \parallel BC$; the multiplier is $\frac{1}{2}$, so $MN = \frac{1}{2}BC$. The whole of Quarter I's midline theorem, in one line of algebra.

Journeys and velocities

Displacements add as vectors, so a journey with several stages is one sum:

$$\text{walk } (3; 0), \text{ then } (0; 4), \text{ then } (-1; 2) \implies \text{total } (2; 6)$$

The total **displacement** is $(2; 6)$, of length $\sqrt{40} \approx 6.3$, while the **distance walked** is $3 + 4 + \sqrt{5} \approx 9.2$. They are different quantities and questions ask for both.

Velocities add in the same way. A boat's velocity through the water plus the current's velocity gives the velocity over the ground — which is why a boat aiming straight across a river never arrives straight across.

02 Worked examples

EXAMPLE 1 $OA = (1; 4)$, $OB = (7; 2)$. Find AB , the midpoint of AB , and the point one third of the way from A to B .

① $AB = OB - OA = (6; -2)$
End minus start.

② $OM = \frac{1}{2}((1; 4) + (7; 2)) = (4; 3)$
Average the position vectors.

③ $OP = OA + \frac{1}{3}AB$

④ $= (1; 4) + (2; -\frac{2}{3}) = (3; 3\frac{1}{3})$

ANSWER

$$AB = (6; -2), M(4; 3), P(3; 3\frac{1}{3})$$

EXAMPLE 2 A cyclist rides 6 km east, then 8 km north. Find the total distance and the displacement.

① $(6; 0) + (0; 8) = (6; 8)$

Add the two displacements.

② $| (6; 8) | = 10 \text{ km}$

The displacement.

③ $6 + 8 = 14 \text{ km}$

The distance actually ridden.

④ The two differ because the route bends.

ANSWER

Distance 14 km, displacement 10 km north-east

EXAMPLE 3 Use vectors to prove that the diagonals of a parallelogram bisect each other.

① $AB = \mathbf{a}, AD = \mathbf{b}$

Name two sides from one vertex.

② $AC = \mathbf{a} + \mathbf{b}$

The diagonal from A.

③ Midpoint of AC: $\frac{1}{2}(\mathbf{a} + \mathbf{b})$ from A.

④ Midpoint of BD: $\mathbf{a} + \frac{1}{2}(\mathbf{b} - \mathbf{a}) = \frac{1}{2}(\mathbf{a} + \mathbf{b})$.

The same point.

⑤ The diagonals cross at their common midpoint.

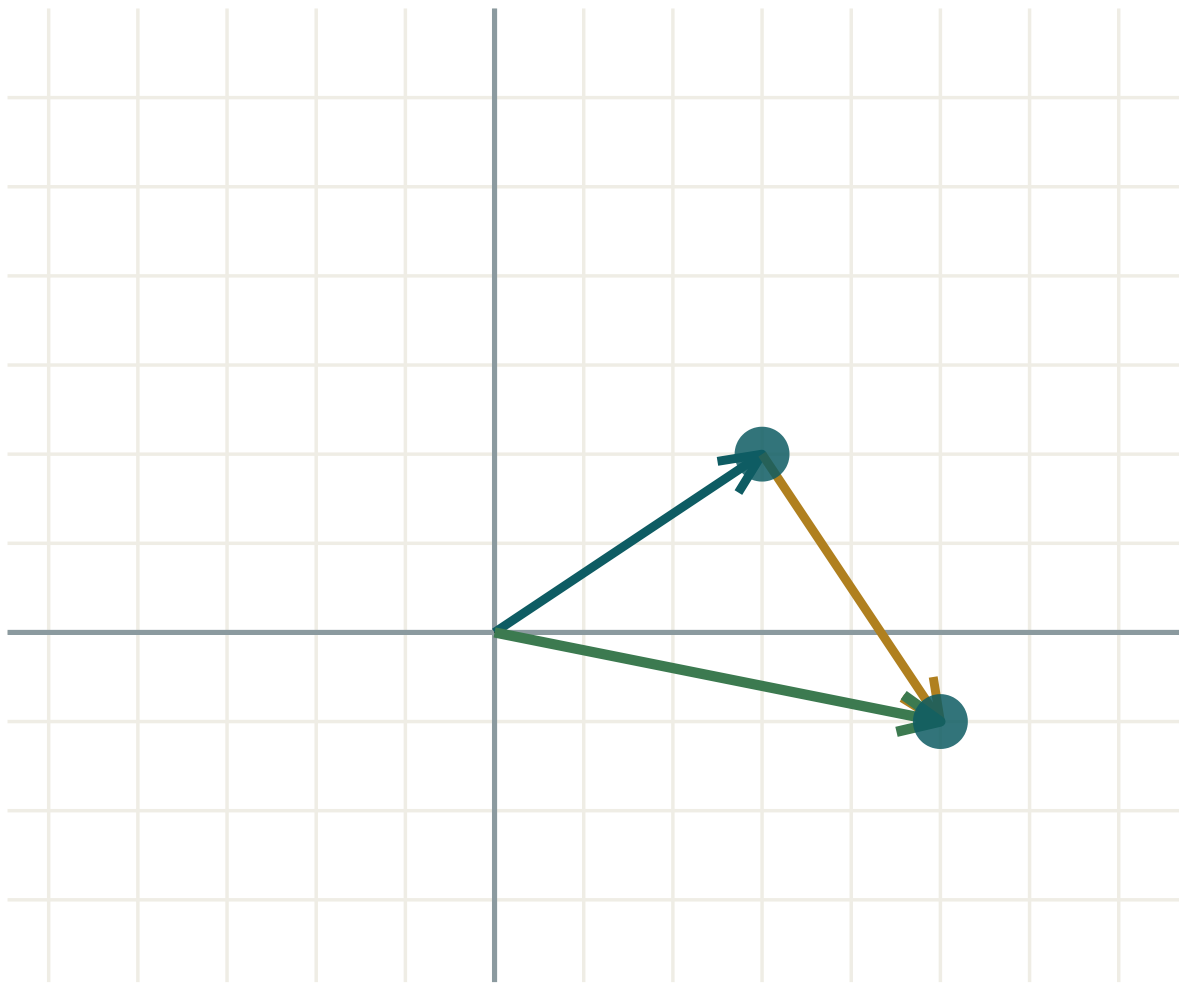
ANSWER

Proved

03 Interactive model

Build a journey out of two vectors and read the resultant off the green arrow.

Two stages of a journey



a (3; 2)

b (2; -3)

$a + b$ (5; -1)

$|a|$ 3.61

$|b|$ 3.61

$|a + b|$ 5.1

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Position vector	Radius-vektor	Радиус-вектор
Resultant displacement	Natijaviy siljish	Результирующее перемещение
Velocity	Tezlik (vektor)	Скорость (вектор)
Component	Tashkil etuvchi	Составляющая
Midpoint (vector form)	O'rta nuqta (vektor shakli)	Середина (в векторах)
Divides in the ratio	Nisbatda bo'ladi	Делит в отношении
Vector proof	Vektorli isbot	Векторное доказательство
Centroid	Medianalar kesishish nuqtasi	Центроид (точка пересечения медиан)
Bearing	Azimut	Азимут

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

If $OA = (2; 1)$ and $OB = (8; 5)$ then AB is:

The midpoint of $A(1; 3)$ and $B(7; 9)$ has position vector:

Walk $(4; 0)$ then $(0; 3)$. The displacement is:

If $MN = \frac{1}{2} BC$ then MN is:

perpendicular to BC

parallel to BC and half its length

equal to BC

unrelated

The distance walked and the displacement are equal only when:

always

the route is a straight line in one direction

the route is closed

never

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $OA = (1; 1)$, $OB = (4; 5)$. Find AB .

ANSWER $(3; 4)$

2. Find the midpoint of $A(0; 0)$ and $B(6; 10)$.

ANSWER $(3; 5)$

3. Walk $(3; 0)$ then $(0; 4)$. Find the displacement.

ANSWER 5

4. Find $2 \cdot (1; 3) + (4; -2)$.

ANSWER $(6; 4)$

5. $OA = (5; 2)$. Write the coordinates of A .

ANSWER $A(5; 2)$

6. Walk $(2; 0)$ then $(-2; 0)$. Find the displacement.

ANSWER 0

7. Find $|(9; 12)|$.

ANSWER 15

Medium · the standard question

7 PROBLEMS

1. $OA = (-1; 3)$, $OB = (5; -1)$. Find AB and its length.

ANSWER $(6; -4)$, $\sqrt{52} = 2\sqrt{13}$

2. Find the midpoint of $A(-3; 4)$ and $B(9; -2)$.

ANSWER $(3; 1)$

3. A cyclist rides 5 km north then 12 km east. Find distance and displacement.

ANSWER 17 km and 13 km

4. Find $3 \cdot (2; -1) - 2 \cdot (4; 1)$.

ANSWER $(-2; -5)$

5. $OA = (2; 2)$, $AB = (6; 4)$. Find OB .

ANSWER $(8; 6)$

6. Find the point one quarter of the way from $A(0; 0)$ to $B(8; 12)$.

ANSWER $(2; 3)$

7. A plane flies at $(0; 300)$ in a wind of $(40; 0)$. Find its ground speed.

ANSWER $\sqrt{91600} \approx 303 \text{ km/h}$

Hard · stretch and reason

7 PROBLEMS

1. $OA = (1; 2)$, $OB = (7; 5)$, $OC = (4; 11)$. Find the centroid of triangle ABC .

ANSWER $(4; 6)$ — the average of the three position vectors

2. $A(1; 1)$, $B(9; 5)$. Find P on AB with $AP : PB = 3 : 1$.

ANSWER $P(7; 4)$

3. Prove with vectors that the midpoints of the sides of a quadrilateral form a parallelogram.

ANSWER With position vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, $\mathbf{d}$, the midpoint figure has opposite sides $\frac{1}{2}(\mathbf{c} - \mathbf{a})$ twice over — equal vectors, so a parallelogram.

4. A boat can do 5 km/h. It heads north across a river flowing east at 3 km/h. Find its speed and how far downstream it lands after crossing 400 m.

ANSWER $\sqrt{34} \approx 5.83$ km/h; it drifts 240 m downstream

5. Walk $(4; 3)$, then $(-1; 5)$, then $(-3; -8)$. Find the total displacement.

ANSWER $(0; 0)$ — the walk returns to its start

6. $OA = (2; 3)$ and $M(5; 7)$ is the midpoint of AB . Find OB .

ANSWER $(8; 11)$

7. Prove with vectors that a quadrilateral with one pair of opposite sides equal and parallel is a parallelogram.

ANSWER If $AB = DC$ as vectors then $AD = AB + BD = DC + BD = BC$, so the other pair is equal and parallel too.

07 Homework

Homework — 6 problems

Geometry 8, Темы 40–41, pp. 90–95. Name your two base vectors before starting any proof.

- $OA = (3; -2)$, $OB = (9; 6)$. Find AB , $|AB|$ and the midpoint of AB .
- Find $4 \cdot (1; -2) - 3 \cdot (2; 1)$.

3. *A walker goes 8 km east then 6 km north. Find distance and displacement.*
4. *$A(2; 1)$, $B(10; 9)$. Find P on AB with $AP : PB = 1 : 3$.*
5. *$OA = (1; 5)$ and $M(4; 2)$ is the midpoint of AB . Find OB .*
6. *Use vectors to show that the midline of a triangle is parallel to the third side.*